MTH 201 Power Series & Taylor's Theorem

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Consider the function $f(x)=\dfrac{1}{1+x}$.

a) Find a power series for $f(x)$ and determine its radius of convergence.

b) Use your answer from part (a) to determine what function the power series $\sum_{n=1}^{\infty} n(-1)^n x^{n-1}$ models and with what radius of convergence.

c) Determine the value of $\sum_{n=1}^{\infty} n(-1)^n \left(\dfrac{3}{4}\right)^{n-1}$. Can you determine the value of $\sum_{n=1}^{\infty} n(-1)^n (2)^{n-1}$? Why or why not?

d) Find a power series centered at $x = 8$ for $f(x) = \frac{1}{1+x}$ and determine its radius of convergence.

e) Which power series (the one from part (a) or part (d)) can be used to create a series that is equal to $f(10)$?

2. Consider the function $f(x) = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}$.

a) Determine what function $f(x)$ resembles and on what interval it resembles this function.

b) Use your answer from part (a) to help determine the exact value of $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} 1/5}{5^n (n+1)}$.

3. Consider the function $f(x) = \sqrt{x}$.

a) Approximate $f(x)$ by a Taylor polynomial with degree 2 centered at $x = 4$.

b) Use the polynomial from part (a) to estimate the value of $\sqrt{4.1}$.

c) Use Taylor's Theorem to estimate the accuracy of the approximation you made in part (b).

d) Use Taylor's Theorem to estimate the accuracy of the approximation when x is within the interval $[4, 4.2]$.

4. Suppose we construct a degree 1 Taylor polynomial centered at $x = 3$ for the function $f(x) = \sqrt{1+x}$. What is the maximum possible error of this polynomial when estimating the value of $\sqrt{5}$?

5. Suppose you have a mystery function $f(x)$, but you know the following

$$f(1)=2, \quad f'(1)=0, \quad f''(1)=5, \quad f^{(3)}(1)=1, \quad f^{(4)}(x)=\frac{1}{\sqrt{x}5^{x}}$$

Estimate $f(2)$ to at least $\frac{1}{100}$ accuracy.

6. Suppose we construct a Taylor series centered at $x = \pi$ for the function $f(x) = \sin(x)$ and we want to use this Taylor series to estimate the value of $\sin(6)$ to 4 decimal places of accuracy. How many terms of the Taylor series would we need to use to accomplish level of precision in our estimation?